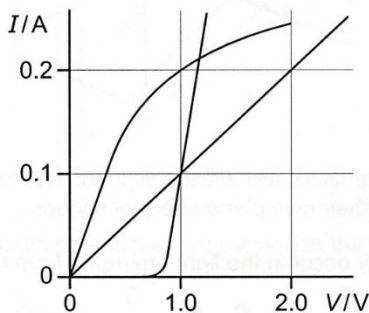


22

D



$$R = \frac{V}{I}$$

for resistor, resistance = $10 \, \Omega$ (constant)

for filament lamp, resistance = $5.0 \, \Omega$ at $1.0 \, \text{V}$

At about $1.2 \, \text{V}$, the graphs for diode and filament lamp intersect and they both have the same I , V values and hence same resistance.

23

A

Before the negative ions of mass m and charge $-e$ enter the metal container, its